PH 3120 – Computational Physics Laboratory-I

Department of Physics
University of Colombo

Self-Learning

Self-learning is the acquisition of knowledge, skills, and understanding without formal instruction or external guidance.

It allows individuals to learn at their own pace and allocate more time to challenging concepts.

Self-learning provides a variety of resources to choose from, such as textbooks, online courses, video tutorials, and forums.

It offers flexibility in receiving feedback through multiple channels, including lecturers, friends, and other experts.

Self-learning enhances the learning process and helps identify areas for improvement.

Why Guidance is important in Self-Learning?

Clear Learning Objectives: Guidance establishes clear learning objectives and goals for learners.

Roadmap: It provides a roadmap for learners, outlining what they need to learn, skills to develop, and milestones to achieve.

Focus and Motivation: Clear objectives help learners stay focused, motivated, and on track throughout their self-learning journey.

Structured Learning Path: Guidance assists in structuring the learning path for learners.

Identification of Resources: It helps learners identify the necessary resources, materials, and activities for effective knowledge acquisition.

Essential Topics: A structured learning path ensures coverage of essential topics and building a strong foundation.

Progression: It facilitates systematic progression in learning, ensuring gradual skill development and knowledge acquisition.

How do we guide you?

Learning the concepts of computational techniques

Developing programs based on the concepts

Applying the program to solve real-world/mathematical problems

How do we guide you?

Learning the concepts of computational techniques

In most of the questions, first part shows you the concepts you must study

1. **First Derivatives**

I. Answer the following questions based on first order numerical differentiation.

(a) Write down the Taylor series for an arbitrary function ($f(x)$) at $x = a$.

(b) Using the Taylor series expansion, derive the finite difference formulars for the first derivative. The three techniques that must be considered are

1. Forward difference
2. Backward difference
3. Central difference

(c) What is the truncation error in context of Taylor series approximation?

(d) Explain the truncation error in the above three methods. Which method has the lowest truncation error?

(e) Compare the forward, backward, and central difference methods for numerical differentiation. Discuss their advantages, disadvantages, and suitable applications for each method.

(f) Why is the choice of step size important in numerical differentiation methods? How does the step size affect the accuracy and stability of the derivative approximation obtained using the above three methods?

How do we guide you?

Developing programs based on the concepts

In most of the questions, second part let you to apply the concepts to develop a program

II. Develop a python function with the following signature.

```python
def num_diff(f, x, h, method):
    """
    Compute the numerical derivative of a function using different difference methods.

    Parameters:
        f (function): The function for which to compute the derivative.
        x (float): The point at which to compute the derivative.
        h (float): The step size for the difference approximation.
        method (str): The differentiation method to use ('forward', 'backward', or 'central').

    Returns:
        float: The estimated derivative value.

    """
```

How do we guide you?

In most of the questions, third part let you to apply the computational technique to solve the real-world problems.

Applying the program to solve real-world/mathematical problems

III. In simple harmonic motion, the position of the object can be described by the equation:

$$x(t) = A\,cos(\omega t + \varphi)$$

Where:

- $x(t)$ represents the position of the object at time t.
- A is the amplitude of the motion.
- ω is the angular frequency, given by ω = 2πf, where f is the frequency of the motion.
- φ is the phase constant that determines the initial position and velocity of the object.

Consider that an object undergoing simple harmonic motion with $\omega = 100$ rad s^{-1}, $A = 2$ cm, and $\phi = \frac{\pi}{6}$.

(a) Plot the position within the interval $-4s \leq t \leq 4s$ with 0.001s step size.

(b) Use the above central_diff() function to calculate the speed $(v(t))$ of the object as a function of time within $-4s \leq t \leq 4s$ interval and plot $v(t)$.

(c) Use the above central_diff() function to calculate the acceleration $(a(t))$ of the object as a function of time within $-4s \leq t \leq 4s$ interval and plot $a(t)$.

You will learn Numerical Methods during Lab-2 to Lab-7

We will cover most of these topics

https://pythonnumericalmethods.berkeley.edu/notebooks/Index.html

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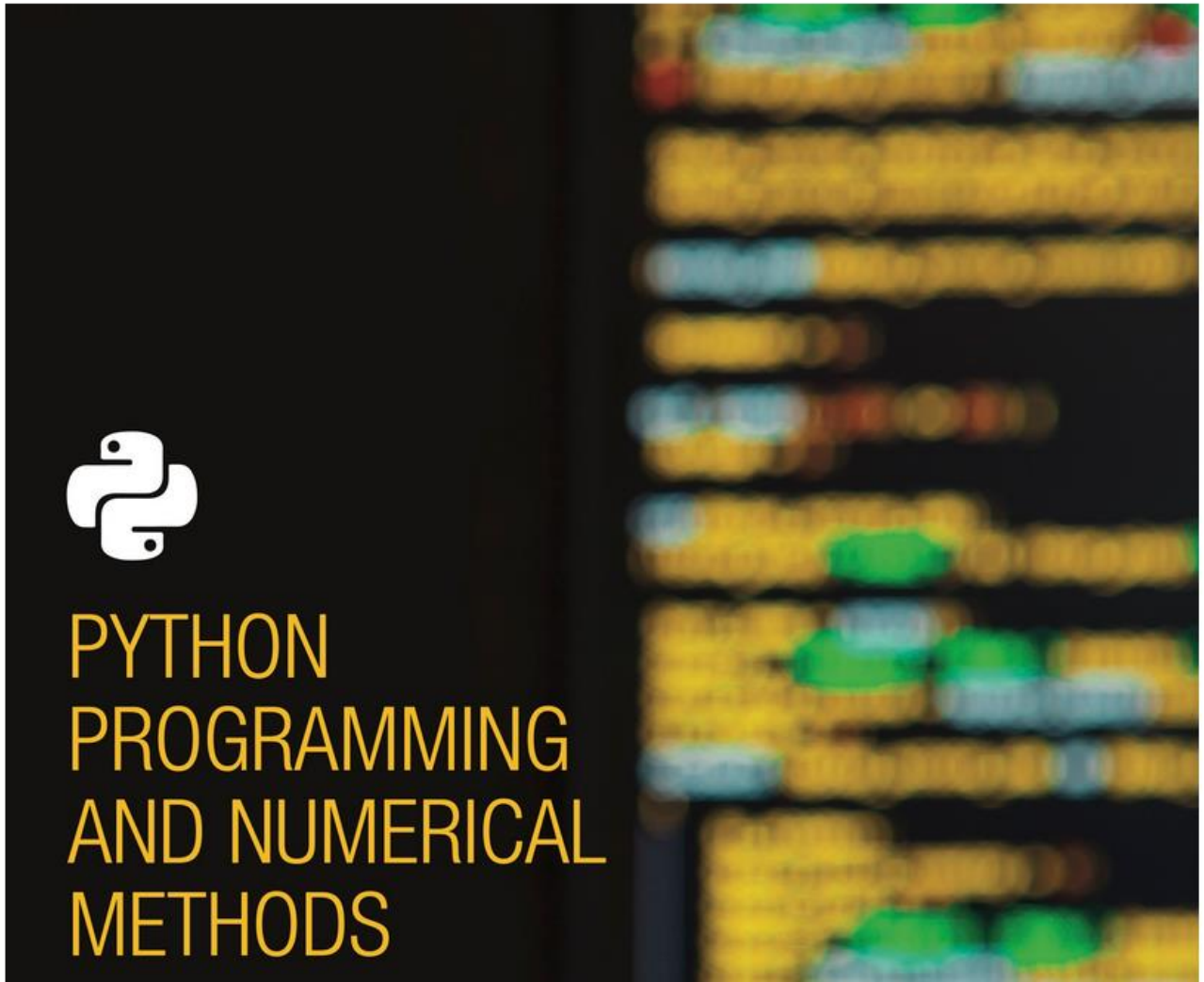
Chapter 8. Complexity
Chapter 9. Representation of Numbers
Chapter 10. Errors, Good Programming Practices, and Debugging
Chapter 11. Reading and Writing Data
Chapter 12. Visualization and Plotting
Chapter 13. Parallel Your Python
Chapter 14. Linear Algebra and Systems of Linear Equations
Chapter 15. Eigenvalues and Eigenvectors
Chapter 16. Least Squares Regression
Chapter 17. Interpolation
Chapter 18. Series
Chapter 19. Root Finding
Chapter 20. Numerical Differentiation
Chapter 21. Numerical Integration
Chapter 22. Ordinary Differential Equation - Initial Value Problems
Chapter 23. Ordinary Differential Equation - Boundary Value Problems
Chapter 24. Fourier Transform
Chapter 25. Introduction to Machine Learning
Appendix A. Getting Started with Python on Windows

Python Programming And Numerical Methods: A Guide For Engineers And Scientists